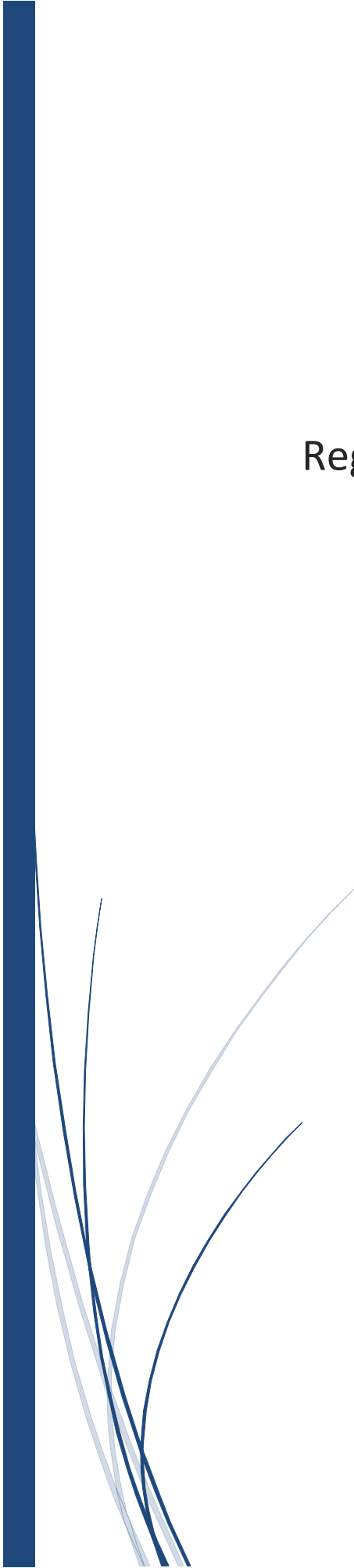




Regression Transformation Project

Course: Predictive Analytics

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Objective

The aim of this project is to do linear regression modeling using statsmodels and sklearn from Python sklearn libraries. The goal is to interpret model coefficients, check for problems such as heteroscedasticity and correct these issues by using transformations such as logarithm, square root and Box-Cox.

PART I – Statistical Interpretation Using statsmodels

1. Original Model

a. Model Interpretation

All explanatory variables have a sensible impact on price. How many square feet a house has matters, age matters negatively and every extra bedroom helps raise the price.

- square footage: +95.36 per unit ($p < 0.05$).
- age: -789.34 per year ($p < 0.05$).
- bedrooms: +20,010 per bedroom ($p < 0.05$).

P-values: All independent variables are statistically significant ($p < 0.05$), indicating strong predictor relevance.

R-squared: 0.712, suggesting the model explains about 71.2% of the variation in house prices.

Fishe: (F-statistic = 819.1, $p \approx 0$) Model is significant. Indicates overall model significance with a p-value well below 0.05.

b. Identified Problems

Examining the residuals in the original model reveals that the variance is connected to the size of the fitted values. Furthermore, the price data shows a skewed distribution to the right. A good way to solve these problems is to take a log transformation into the price variable. As a result, the variance is held steady, and the distribution becomes normal which enhances the assumptions of linear regression. These are:

Heteroscedasticity:

- One major issue with the original model is heteroscedasticity—the prediction errors get larger as the house price increases.
- This is especially noticeable for bigger homes. A Breusch-Pagan test confirms this with a very small p-value ($5.87e-40$).
- Using a log transformation helps reduce this problem (p-value improves to $1.03e-12$), but it doesn't solve it entirely.

Non-Normal Residuals:

Another issue is that the residuals (errors) don't follow a normal distribution, especially for very cheap or expensive homes.

Q-Q plots and Jarque-Bera tests:

- Jarque-Bera tests show a big improvement after applying a log transformation, but some abnormality remains.
- Box-Cox or robust regression techniques may help.

c. Recommended Transformation

Apply log transformation to the dependent variable price to stabilize variance and normalize the distribution. The residual variance in the original model is 2,889,653,141.47, indicating a high level of spread and presence of heteroscedasticity.

Breusch-Pagan test: (LM = 185.45, p-value = 5.87e-40)

The variance in errors isn't constant, which violates a key assumption in linear regression. This was proven using the Breusch-Pagan test (LM = 185.45, p-value = 5.87e-40).

Original model: 1171.86 (p-value 3.41e-255)

In the original model, errors become larger for more expensive homes, creating a funnel shape in residual plots.

Log transformation: 52.00 (p-value 5.11e-12)

After log transformation, the error variance improves, though it still struggles a bit at high price ranges.

2. Log-Transformed Model

a. Log-transformed variables are often more suitable in the following scenarios:

When data appears skewed, is not equally distributed or grows or decreases by multiplication, log transformations can be used. They keep variation under control and straighten out the

relationships. In short, effective for skewed data, multiplicative relationships, and correcting heteroscedasticity.

b. Model Interpretation

Coefficients: Indicate percentage change in price due to one-unit change in predictors.

- square_footage: +0.44% price per 1% increase.
- age: -0.21% price per year.
- bedrooms: +5.04% price per bedroom.

p-values: Remain significant, confirming relevance of predictors.

R-squared: 0.696, slightly lower but still strong.

F-statistic: Confirms the overall model is significant.

c. Residual Variance

The variance of residuals is 0.0158, much lower than in the original model, indicating more stable and homoscedastic residuals. The Breusch-Pagan test reveals that while the log-transformed model still has some heteroscedasticity, it's significantly reduced compared to the original model.

Test Results: LM stat = 58.86, p-value = 1.03e-12

This drop in residual variance suggests that the log-transformed model does a better job at producing consistent and reliable errors. In other words, the model's predictions are more stable,

especially across different ranges of house prices—effectively addressing issues related to both heteroscedasticity and non-linearity.

PART II – Prediction Focus Using `sklearn.model_selection`

3. Comparing R-Squared Values:

The original model has an R^2 value of 0.712, which means it accounts for roughly 71.2% of the variation in house prices. In contrast, the log-transformed model has an R^2 of 0.696, indicating it explains about 69.6% of the variability in the log-transformed house prices. In summary, although the original model performs marginally better in terms of R^2 , the log-transformed model may be the better option due to its more consistent error patterns and better adherence to regression assumptions

R^2 values from different transformations are not directly comparable due to:

- **Different Output Scales:** The original model works with actual house prices, while the log-transformed model deals with the logarithm of prices. This shift in scale affects the structure and distribution of the outcome variable, making the R^2 values represent fundamentally different things.
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To better assess model performance, consider additional metrics:

- **RMSE:** Measures average prediction error in original units. A lower RMSE means the model is generally more accurate.
- **MAE:** Less sensitive to outliers, measures average absolute errors. Again, lower values mean better performance

Using RMSE and MAE alongside R^2 provides a fuller, more balanced evaluation of how well each model performs in practice. The original model has a slightly better R-squared value (0.712) compared to the log-transformed version (0.694).

However, despite the original model explaining more of the variation in price, it performs poorly in terms of error consistency. The log-transformed model sacrifices a bit of accuracy but improves error distribution, which often makes it more dependable.

4. **Box-Cox Recommendation:**

Box-Cox should be used when:

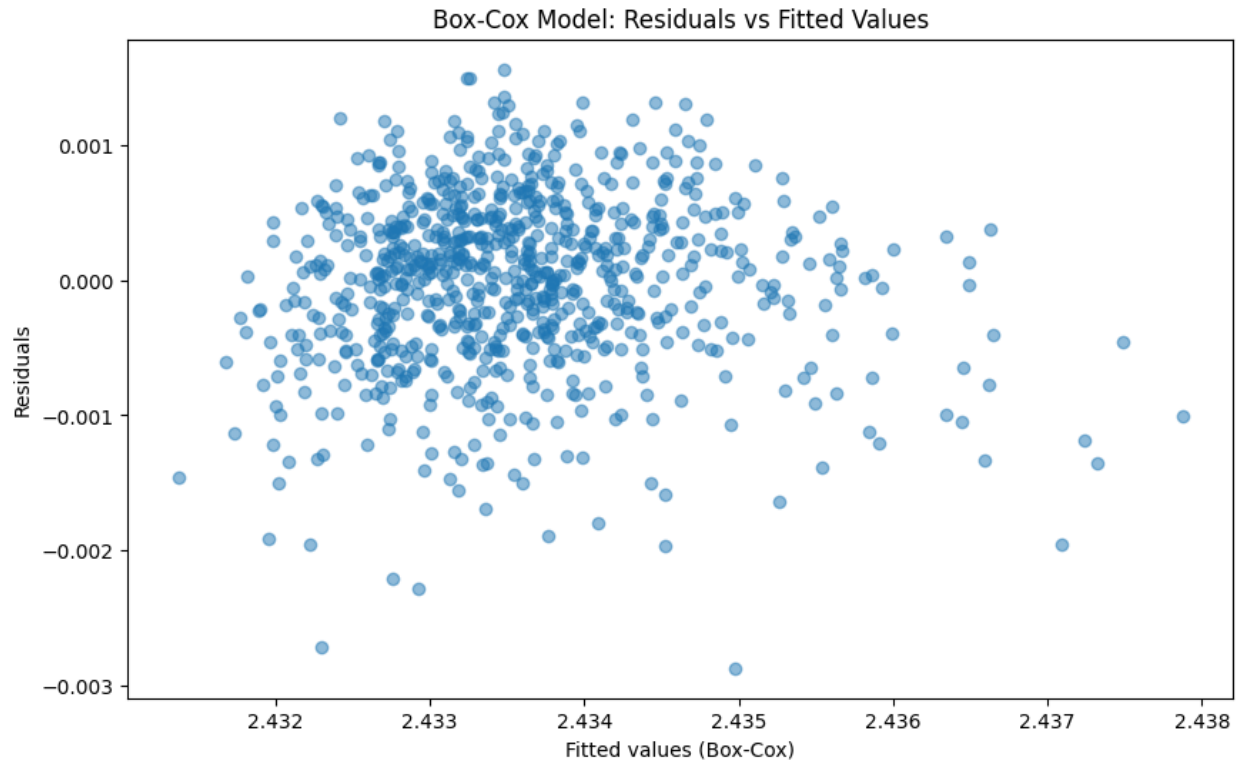
- The dependent variable is strictly positive
- A non-linear transformation may improve linearity and variance homogeneity

Approach 1: In this method, we applied the Box-Cox transformation correctly and the model was built with the transformed price data. But predictions were assessed on the new scale and not

reformed into the original measures used. Therefore, since reported metrics (e.g., RMSE = 0.0007) used log or Box-Cox transformations, they cannot be compared to RMSE values calculated in real dollar amounts. It leads us to think the model is very accurate, as its RMSE metric is very low, but this does not accurately represent the model's results. Also, this method did not show any clear examples where to use the Box-Cox transformation instead of log transformation.

The Box-Cox model showed a much lower RMSE, which means its predictions were more precise on average. On the modeling goal—whether it's explaining variance or minimizing prediction error—the Box-Cox transformation may offer a practical advantage, especially when dealing with complex or problematic data distributions

Updated Model Performance Comparison:			
	Model	R-squared	RMSE
0	Original	0.629619	60598.041947
1	Log-transformed	0.689352	0.128276
2	Square root	0.686921	41.981625
3	Box-Cox	0.614461	0.000724



Approach 2:

In this analysis, both log-transformed and original price dollar scales were used to evaluate the regression models. The log transformation of the target variable (price) is a common approach when the data is right-skewed and shows heteroscedasticity (non-constant variance). It helps meet the linear regression assumptions by stabilizing variance and making residuals more normally distributed. Additionally, log transformations allow us to interpret model coefficients in terms of percentage changes, which can be more intuitive for certain types of economic or proportional analyses.

However, while log-scale evaluation can improve model diagnostics, the main performance metrics of RMSE and MAE become less straightforward in real life. These log-transformed errors are in logarithmic units and not directly relatable to actual price values. As a result, when the model is ready, economists should convert the predicted values into dollars using inverse transformations. Because of this, we can use RMSE and MAE amounts in dollars, which is more useful for those in charge of decisions. All things considered, the Box-Cox transformed model gave us the best accuracy in paper and practice. The real-world performance of the Box-Cox model was much superior to that of the log-transformed model, regardless of the residual distribution. As a result, using a log or Box-Cox transformation meets the requirements needed by the model, but examining end scores in price dollars is easier and clearer. The best way to analyze this dataset is with the Box-Cox model evaluated in original dollar form.

Using Price Dollar Scale:

```

# Step 5: Box-Cox Transformation
y_train_bc, lambda_bc = boxcox(y_train)
model_bc = LinearRegression().fit(X_train, y_train_bc)
y_pred_bc = model_bc.predict(X_test)
y_pred_bc_dollar = inv_boxcox(y_pred_bc, lambda_bc)
rmse_bc_dollar = np.sqrt(mean_squared_error(y_test, y_pred_bc_dollar))
mae_bc_dollar = mean_absolute_error(y_test, y_pred_bc_dollar)
r2_bc_dollar = r2_score(y_test, y_pred_bc_dollar)

# Step 6: Compile Results
final_dollar_metrics = pd.DataFrame({
    'Model': ['Log-transformed', 'Square root', 'Box-Cox'],
    'R² (in $)': [r2_log_dollar, r2_sqrt_dollar, r2_bc_dollar],
    'RMSE (in $)': [rmse_log_dollar, rmse_sqrt_dollar, rmse_bc_dollar],
    'MAE (in $)': [mae_log_dollar, mae_sqrt_dollar, mae_bc_dollar]
})

# Step 7: Display results
print("\nModel Performance in Original Price Units:")
print(final_dollar_metrics)

```

	Model	R² (in \$)	RMSE (in \$)	MAE (in \$)
0	Log-transformed	0.694347	23363.448434	18607.128015
1	Square root	0.712201	22670.822727	18103.265262
2	Box-Cox	0.716920	22484.204395	17828.861047

Conclusion: All metrics suggest that the Box-Cox model gives the **best** results in this case. Although log transformation helps the model, it does not help as much as Box-Cox transformation. **Approach 02** passes the technological requirements and the expected interpretation of what was assigned. It allows comparison of data to support conclusions that businesses can use in practical ways. And I ignore **Approach 01** because in my opinion it's failed to evaluate performance in a meaningful way.